NOTES: (UNLESS OTHERWISE SPECIFIED)

1. ABSIS TYPE-A CABLE (RS485 BUS CABLE)
 - 1.1. PN: ABSIS-CA-A-XXX (WHERE "XXX" IS WIRE LENGTH IN MM)
 - 1.2. CONNECTORS: 2X 39-01-3065 (5557-06R-BL)
 - 1.3. POWER/GND CONTACTS: 8X 39-00-0211
 - 1.4. DATA CONTACTS: 4X 39-00-0046
 - 1.5. DATA CONDUCTORS A & B: 2X 22AWG STRANDED COPPER CONDUCTORS, TWISTED AS A SHIELDED PAIR (STP), INSULATION COLOR AS DESIRED.
 - 1.6. POWER/GND CONDUCTORS: 4X 16AWG STRANDED COPPER CONDUCTORS, WHITE INSULATION.
 - 1.7. MATES WITH: TYPE-E
2. ABSIS TYPE-B CABLE (BL BUS CABLE)
 - 2.1. PN: ABSIS-CA-B-XXX (WHERE "XXX" IS WIRE LENGTH IN MM)
 - 2.2. CONNECTORS: 2X 39-01-3045 (5557-04R-BL)
 - 2.3. CONTACTS: 8X 39-00-0211
 - 2.4. CONDUCTORS: 4X 16AWG STRANDED COPPER CONDUCTORS, WHITE INSULATION.
 - 2.5. MATES WITH: TYPE-F
3. ABSIS TYPE-C CABLE (I/O POWER BUS CABLE)
 - 3.1. PN: ABSIS-CA-C-XXX (WHERE "XXX" IS WIRE LENGTH IN MM)
 - 3.2. CONNECTORS: 2X 39-01-3085 (5557-08R-BL)
 - 3.3. CONTACTS: 16X 39-00-0211
 - 3.4. CONDUCTORS: 8X 16AWG STRANDED COPPER CONDUCTORS, WHITE INSULATION.
 - 3.5. MATES WITH: TYPE-G
4. ABSIS TYPE-D CABLE (BL POWER BUS CABLE)
 - 4.1. PN: ABSIS-CA-D-XXX (WHERE "XXX" IS WIRE LENGTH IN MM)
 - 4.2. CONNECTORS: 2X 39-01-3105 (5557-10R-BL)
 - 4.3. CONTACTS: 20X 39-00-0211
 - 4.4. CONDUCTORS: 10X 16AWG STRANDED COPPER CONDUCTORS, WHITE INSULATION.
 - 4.5. MATES WITH: 39-01-2101 (5559-10P)
 - 4.6. MATING CONTACTS: 10X 39-00-0081
5. ABSIS TYPE-E CABLE (RS485 BUS BULKHEAD CABLE)
 - 5.1. PN: ABSIS-CA-E-XXX (WHERE "XXX" IS WIRE LENGTH IN MM)
 - 5.2. CONNECTORS: 2X 39-01-2061 (5559-06P)
 - 5.3. POWER/GND CONTACTS: 8X 39-00-0081
 - 5.4. DATA CONTACTS: 4X 39-00-0048
 - 5.5. DATA CONDUCTORS A & B: 2X 22AWG STRANDED COPPER CONDUCTORS, TWISTED AS A SHIELDED PAIR (STP), INSULATION COLOR AS DESIRED.
 - 5.6. POWER/GND CONDUCTORS: 4X 16AWG STRANDED COPPER CONDUCTORS, WHITE INSULATION.
 - 5.7. LENGTH: SHORT, PIGTAIL
 - 5.8. MATES WITH: TYPE-A
6. ABSIS TYPE-F CABLE (BL BUS BULKHEAD CABLE)
 - 6.1. PN: ABSIS-CA-F-XXX (WHERE "XXX" IS WIRE LENGTH IN MM)
 - 6.2. CONNECTORS: 2X 39-01-2041 (5559-04P)
 - 6.3. CONTACTS: 8X 39-00-0081
 - 6.4. CONDUCTORS: 4X 16AWG STRANDED COPPER CONDUCTORS, WHITE INSULATION.
 - 6.5. MATES WITH: TYPE-B
7. ABSIS TYPE-G CABLE (I/O POWER BUS BULKHEAD CABLE)
 - 7.1. PN: ABSIS-CA-G-XXX (WHERE "XXX" IS WIRE LENGTH IN MM)
 - 7.2. CONNECTORS: 2X 39-01-2081 (5559-08P)
 - 7.3. CONTACTS: 16X 39-00-0081
 - 7.4. CONDUCTORS: 8X 16AWG STRANDED COPPER CONDUCTORS, WHITE INSULATION.
 - 7.5. MATES WITH: TYPE-C
8. ABSIS TYPE-H CABLE (I/O ACCESSORY BULKHEAD CABLE)
 - 8.1. PN: ABSIS-CA-H-XXX (WHERE "XXX" IS WIRE LENGTH IN MM)
 - 8.2. CONNECTOR 1: 1X 39-01-2221 (5559-22P)
 - 8.3. CONNECTOR 1 CONTACTS: 22X 39-00-0048
 - 8.4. CONNECTOR X: ALL OTHER CONNECTORS PER PRINT.
 - 8.5. CONNECTOR X CONTACTS: PER PRINT.
 - 8.6. CONDUCTORS: 22X 26AWG STRANDED COPPER CONDUCTORS, WHITE INSULATION.
 - 8.7. MATES WITH: 39-01-2220
 - 8.8. MATING CONTACTS: 22X 39-00-0209
9. ABSIS TYPE-J CABLE (PWR ACCESSORY CABLE)
 - 9.1. PN: ABSIS-CA-J-XXX (WHERE "XXX" IS WIRE LENGTH IN MM)
 - 9.2. CONNECTOR 1: 1X 39-01-3025 (5557-02R-BL)
 - 9.3. CONNECTOR 1 CONTACTS: 2X 39-00-0211
 - 9.4. CONNECTOR X: ALL OTHER CONNECTORS PER PRINT.
 - 9.5. CONNECTOR X CONTACTS: PER PRINT.
 - 9.6. CONDUCTORS: 2X 16AWG STRANDED COPPER CONDUCTORS, WHITE INSULATION.
10. GENERAL CABLE ASSEMBLY REQUIREMENTS:
 - 10.1. FABRICATE CABLES TO THE CONNECTOR, CONTACT, CONDUCTOR SIZE, AND PINOUT REQUIREMENTS SHOWN ON THIS DRAWING.
 - 10.2. USE STRANDED COPPER CONDUCTORS. CONDUCTOR INSULATION SHALL BE UNDAMAGED AND SUITABLE FOR THE INTENDED TERMINAL AND HARNESS APPLICATION.
 - 10.3. CRIMP TERMINALS USING A TOOL SUITABLE FOR THE SPECIFIED CONTACT AND CONDUCTOR SIZE. DO NOT SOLDER CRIMP TERMINALS UNLESS SPECIFICALLY DIRECTED.
 - 10.4. TERMINALS SHALL BE FULLY SEATED AND RETAINED IN THEIR CONNECTOR HOUSINGS. PERFORM A LIGHT PULL CHECK AFTER ASSEMBLY.
 - 10.5. PROVIDE ADEQUATE STRAIN RELIEF AT CONNECTORS, SPLICES, AND CABLE BREAKOUTS. LOADS SHALL NOT BE CARRIED BY INDIVIDUAL TERMINALS OR CONDUCTORS.
 - 10.6. INSULATE ALL SPLICES, UNUSED CONDUCTORS, DRAIN WIRES, AND EXPOSED CONDUCTIVE MATERIAL USING HEAT SHRINK OR EQUIVALENT INSULATION.
 - 10.7. ROUTE AND SECURE CONDUCTORS TO PREVENT CHAFING, SHARP BENDS, EXCESSIVE TENSION, AND INTERFERENCE WITH MOVING PARTS.
 - 10.8. CABLE LENGTH SHALL BE SUFFICIENT FOR INSTALLATION AND SERVICE WITHOUT PLACING TENSION ON CONNECTORS. AVOID EXCESSIVE UNSECURED SLACK.
 - 10.9. IDENTIFY CABLES AND CONNECTORS AS REQUIRED TO SUPPORT INSTALLATION, TROUBLESHOOTING, AND CORRECT MATING.
 - 10.10. VERIFY CONTINUITY, PIN-TO-PIN WIRING, AND ABSENCE OF UNINTENDED SHORTS BEFORE INSTALLATION.
 - 10.11. WHERE REQUIREMENTS IN A CABLE-SPECIFIC NOTE CONFLICT WITH THESE GENERAL REQUIREMENTS, THE CABLE-SPECIFIC REQUIREMENT SHALL GOVERN.
11. RS485 DATA RUN REQUIREMENTS
 - 11.1. DATA CONDUCTORS A AND B SHALL BE STRANDED COPPER CONDUCTORS, TWISTED AS A SHIELDED PAIR (STP). THE SHIELD SHALL NOT BE USED AS BUS GND, POWER RETURN, OR A CURRENT-CARRYING CONDUCTOR.
 - 11.2. MAINTAIN SHIELD CONTINUITY THROUGH THE COMPLETE RS485 DATA RUN, INCLUDING ALL INTERMEDIATE CABLE, BULKHEAD, OR HARNESS CONNECTIONS.
 - 11.3. THE SHIELD SHALL BE BONDED TO GND ONLY AT THE OWNING ABSIS BUS MASTER J6 CONNECTION. DO NOT BOND THE SHIELD TO BUS GND, PCB GND, CHASSIS, OR STRUCTURE AT INTERMEDIATE CONNECTIONS OR AT THE FAR END OF THE DATA RUN.
 - 11.4. THE FAR END OF THE SHIELD SHALL BE INSULATED AND LEFT UNBONDED.
 - 11.5. INSULATE AND STRAIN-RELIEVE ALL SHIELD/DRAIN SPLICES, PIGTAILS, AND TERMINATIONS. KEEP EXPOSED DRAIN CONDUCTORS AND PIGTAILS AS SHORT AS PRACTICAL.
 - 11.6. WHERE A CABLE DRAIN CONDUCTOR IS NOT COMPATIBLE WITH THE REQUIRED CRIMP CONTACT, USE AN INSULATED SPLICE TO A COMPATIBLE STRANDED PIGTAIL. DO NOT FORCE AN OVERSIZED CONDUCTOR OR MULTIPLE DRAIN CONDUCTORS INTO A SINGLE CRIMP CONTACT.
 - 11.7. EACH ABSIS BUS MASTER RS485 SEGMENT SHALL HAVE A SEPARATE DETACHABLE SHIELD-DRAIN CONNECTION ASSOCIATED WITH ITS J2, J3, OR J4 BUS CABLE.
 - 11.8. CONNECT ABSIS BUS MASTER J6 TO A SHORT, SECURED LEAD AND INSULATED FAN-OUT PROVIDING ONE DRAIN CONNECTION FOR EACH RS485 SEGMENT. REMOVAL OF AN RS485 BUS CABLE SHALL NOT REQUIRE LOOSENING OR REMOVING THE J6 SCREW CONNECTION.
 - 11.9. MASTER-SIDE FIXED DRAIN CONNECTOR SHALL USE:
 - 11.9.1. SMP-02V-BC HOUSING
 - 11.9.2. 1X SHF-001T-0.8BS FEMALE CONTACT
 - 11.9.3. POPULATE CAVITY 1 ONLY.
 - 11.9.4. CAVITY 2 SHALL CONTAIN NO CONTACT OR CONDUCTOR.
 - 11.10. REMOVABLE CABLE-SIDE DRAIN CONNECTOR SHALL USE:
 - 11.10.1. SMR-02V-B HOUSING
 - 11.10.2. 1X SYM-001T-P0.6(N) MALE CONTACT
 - 11.10.3. POPULATE CAVITY 1 ONLY.
 - 11.10.4. CAVITY 2 SHALL CONTAIN NO CONTACT OR CONDUCTOR.
 - 11.11. VERIFY CAVITY NUMBERING FROM THE CONNECTOR MATING-FACE VIEW. DO NOT DETERMINE CAVITY NUMBER FROM WIRE-ENTRY ORIENTATION ALONE.
 - 11.12. LABEL BOTH HALVES OF EACH DRAIN CONNECTOR WITH THE ASSOCIATED RS485 SEGMENT AND ROUTE AND SECURE THE DRAIN CONNECTOR WITH ITS ASSOCIATED BUS CABLE.
 - 11.13. THE TWO-POSITION JST SM HOUSING IS NOT MECHANICALLY KEYED TO THIS APPLICATION. ROUTE, RESTRAIN, AND IDENTIFY DRAIN CONNECTORS TO PREVENT CONNECTION TO OTHER TWO-POSITION JST SM CIRCUITS.
 - 11.14. VERIFY CORRECT PIN-TO-PIN CONTINUITY OF ALL RS485 BUS CONDUCTORS AND CONTINUITY OF THE SHIELD THROUGH THE COMPLETE DATA RUN.
 - 11.15. WITH THE MASTER-END DRAIN CONNECTION DISCONNECTED, VERIFY THE SHIELD IS ISOLATED FROM BUS GND, PCB GND, CHASSIS, AND STRUCTURE.
 - 11.16. VERIFY CAVITY 1 IS POPULATED AND CAVITY 2 IS EMPTY ON BOTH HALVES OF EACH TWO-POSITION JST SM DRAIN CONNECTION.
 - 11.17. VERIFY CRIMP-CONTACT RETENTION, INSULATION, STRAIN RELIEF, AND CABLE RESTRAINT.
 - 11.18. AFTER ASSEMBLY, VERIFY NORMAL RS485 COMMUNICATION WITH THE INTENDED BUS MASTER, REPRESENTATIVE NODES, AND REPRESENTATIVE SWITCHED ELECTRICAL LOADS OPERATING.

REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED
-	INITIAL RELEASE	5/5/2023	J. STEENSEN
A	UPDATED TYPES A & E CABLES TO REQUIRE TSP WIRES FOR RS485 LINES.	11/2/2025	J. STEENSEN
B	ADDED SHIELDED TO DATA CONNECTORS FOR TYPE-A/E CABLES REMOVED "MATES WITH TYPE-K" FROM TYPE-J CABLES ADDED NOTES 10 & 11 FOR GENERAL REQUIREMENTS FOR THE CABLES AND SHIELDS	9/19/2026	J. STEENSEN

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES.

DUAL DIMENSIONS ARE IN MILIMETERS AND ENCLOSED IN SQUARE BRACKETS.

INTERPRET DRAWING IAW ASME Y14.100-2017.

INTERPRET DIM AND TOL IAW ASME Y14.5-2009.

REMOVE ALL BURRS AND SHARP EDGES TO .05 MAX.

DIMENSIONS APPLY BEFORE FINISH.

UOS INCH TOLERANCES ARE:

DECIMALS:
.X ± .1
.XX ± .03
.XXX ± .010

ANGLES:
MACHINED ± .5°
LOCATING ± .5°
FORMED ± 2°
CHAMFERS ± 5°

DRILLED HOLE DIAMETERS.
.0135 - .125: +.004/-.001
.126 - .250: +.005/-.001
.251 - .500: +.006/-.001
.501 - .750: +.008/-.001
.751 - 1.000: +.010/-.001
1.001 - 2.000: +.012/-.001

DESIGNED J. STEENSEN 05/05/2023
CHECKED
DRAWN J. STEENSEN 05/05/2023

CC BY NC SA

OPEN HORNET
https://www.OpenHornet.com

TITLE
ABSIS CABLES

SIZE D | DWG NO OH-SPEC-002 | REV B

SCALE N/A | DO NOT SCALE DWG | SHEET 1/1